

Phase 2.4.1 — Quantitative and Diversity Analysis of LCRs: Context, Design, and Findings

Status: analysis complete; all six methods validated at LCR and protein levels; continuous-metric and machine-checked concordance layers complete; findings finalized.

Date: August 2026.

1. Project context

This work is part of the PhD research program on **low-complexity regions (LCRs) in human RNA-binding proteins (RBPs)**, within the broader benchmarking effort across LCR detection tools: **CAST, SEG, fLPS, LCRFinder, and AlcoR**, with SEG additionally run in an intermediate-stringency variant (SEG_intermediate). Earlier phases produced per-method LCR calls, a rule-based physicochemical behavior annotation layer (signatures such as basic_enriched_region, arginine_rich_rna_contact_region, sr_rs_repeat_region), RNA-target superclass assignments, and domain-position classifications.

2. Phase 2.4.1 objective

A first **quantitative and diversity analysis** of LCR annotations, examining how LCR characteristics vary by **RNA-target superclass** (mRNA, tRNA, pre-rRNA, snRNA, snoRNA, ncRNA, ribosomal protein, diverse, unknown), stratified by detection method, across five analysis levels:

- 1. Domain-position enrichment relative to RNA classes
- 2. Physicochemical-property enrichment relative to RNA classes
- 3. Sequence-level properties (length, amino-acid enrichment)
- 4. LCR multiplicity per protein (1 vs. multiple), per method
- 5. Overall residue coverage and physicochemical-property fractions

3. Input data

Input	Role	Key columns
lcr_annotations.xlsx (Sheet1, 20,954 rows)	LCR calls with annotations	protein_id (pipe-delimited composite), source_method, start, end, length, sequence, rna_target_superclass, domain_position_class, primary_physicochemical_annotation
lcr_features.xlsx (3 sheets: composition, distribution, co_occurrence; 20,954 rows each)	Per-LCR measurements	frac_A...frac_Y, property fractions, fcr, ncpr, frac_GS, RG/SR/GS repeat columns, run/gap distribution statistics, cooc_positive_aromatic

Input	Role	Key columns
Master RBP table (Combined sheet)	Protein lengths only	uniprot_accession, uniprot_length

4. Measured metrics and units of measurement

4.1 Identifiers and coordinates (per LCR)

Metric	Unit	Description
uniprot_accession	— (identifier)	UniProt accession parsed from the composite protein ID (text before the first \)
method	— (categorical)	Detection tool; canonical labels: CAST, SEG, SEG_intermediate, FLPS, LCRFinder, AlcoR
start, end	amino-acid position (1-based)	LCR interval boundaries
length	amino acids (aa)	LCR length
rna_target_superclass	— (categorical)	RNA-target superclass of the parent protein (9 classes)
domain_position_class	— (categorical)	domain_intrinsic, domain_edge, domain_adjacent, interdomain_linker, distal_terminal, unclassified_no_pfam
primary_physicochemical_annotation	— (categorical)	Primary physicochemical behavior signature

4.2 Composition metrics (per LCR)

All fraction metrics are unitless ratios in [0, 1].

Metric	Unit	Description
frac_A ... frac_Y	fraction (0–1)	Per-residue amino-acid fractions (20 columns)
frac_polar	fraction (0–1)	Polar residues (R, N, D, Q, E, H, K, S, T, W, Y)
frac_hydrophobic	fraction (0–1)	Hydrophobic residues (A, C, G, H, I, K, L, M, F, T, W, V, Y)
frac_strong_hydro	fraction (0–1)	Strong hydrophobes (V, I, L, M, F, W, Y)
frac_aromatic	fraction (0–1)	Aromatic residues (F, Y, W, H)
frac_disorder	fraction (0–1)	Disorder-promoting residues, TOP-IDP set (A, R, G, Q, S, P, E, K)
frac_positive (f+)	fraction (0–1)	Positively charged at pH 7.4 (R, K)
frac_negative (f-)	fraction (0–1)	Negatively charged at pH 7.4 (D, E)

Metric	Unit	Description
fcr	fraction (0–1)	Fraction of charged residues, f+ + f– (Das & Pappu 2013)
ncpr	unitless (–1 to +1)	Net charge per residue, f+ – f–
frac_GS	fraction (0–1)	Combined glycine + serine fraction

4.3 Repeat metrics (per LCR)

Metric	Unit	Description
n_RG_motifs, n_SR_motifs, n_GS_motifs	count (integer)	Non-overlapping motif matches (RG-like = $\text{RG}\{1, 2\} \setminus \text{RPR}$; RS-like = $\text{RS} \setminus \text{SR}$; GS-like = $\text{GS} \setminus \text{SG}$)
*_repeat_cluster, *_repeat_region	— (boolean)	≥3 motifs with spacing ≤10 aa (Gerstberger et al. 2014)

4.4 Distribution metrics (per LCR; per property track: polar, hydrophobic, aromatic, disorder, charged)

Metric pattern	Unit	Description
<track>_n_runs	count (integer)	Number of contiguous property runs
<track>_mean_run_length, <track>_max_run_length	aa	Run-length statistics
<track>_mean_gap	aa	Mean gap between runs
<track>_cv_gap	unitless	Coefficient of variation of gap lengths
<track>_label	— (categorical)	compact (≤3 runs, max run > 35% of L), dispersed, insufficient

4.5 Co-occurrence metric (per LCR)

Metric	Unit	Description
cooc_positive_aromatic	fraction of windows (0–1)	Fraction of windows of size $\max(5, L // 4)$ containing ≥1 positive and ≥1 aromatic residue (cation– π potential)

4.6 Derived analysis metrics (computed in Phase 2.4.1)

Metric	Unit	Level	Description
coverage_per_lcr	fraction of protein length (0–1)	LCR	length / uniprot_length
n_lcr	count (integer); categorized 1 / 2 / 3+	Protein	LCRs per protein per method

Metric	Unit	Level	Description
lcr_residues	aa	Protein	Σ LCR lengths per protein per method
coverage	fraction of protein length (0–1)	Protein	$\text{lcr_residues} / \text{uniprot_length}$
property_fraction_<signature>	fraction of protein length (0–1)	Protein	Residues in LCRs with primary signature <i>s</i> / uniprot_length
log2_obs_exp	log2 fold ratio (unitless)	Contingency-table cell	0 = expected, +1 = 2× enriched, -1 = 2× depleted
proportion_within_rna_class	fraction (0–1)	Contingency-table cell	Column-normalized share within an RNA class
Summary statistics	source-metric unit	Group	n, median, Q1, Q3, IQR (= Q3 – Q1), mean, SD
p_value / p_fdr	probability (0–1)	Test	Chi-square / Kruskal–Wallis; BH-FDR within analysis block

5. Key design decisions

- **Positional analysis:** five true position classes on Pfam-classified proteins; unclassified_no_pfam analyzed as a separate binary question.
- **Physicochemical analysis:** primary annotations only; both sequence-count and residue-count bases reported.
- description column excluded; measurements from the metrics workbook only.
- **No filtering:** AlcoR retained; no LCRFinder length threshold; overlaps fixed upstream.
- **Pseudo-replication:** every analysis at LCR level and protein level; protein-level significance is the robustness criterion.
- **Method as stratum:** all analyses per method; cross-method comparison via the consolidation layer.

6. Statistical contract

Question type	Measure	Test
Categorical × RNA class	$\log_2(\text{observed/expected})$ per cell	Chi-square
Continuous × RNA class	Median + IQR and mean $\pm$ SD	Kruskal–Wallis

BH-FDR per analysis block, $\alpha = 0.05$; $\log_2$ uniform; effect sizes lead. Visualization: histograms and stacked bars only; fraction histograms grouped into multi-panel figures.

7. Scope limitations

- Analyses are **compositional over LCR-positive proteins** (no LCR-negative background; the master table lacks RNA-class labels).
- Categorical table p-values are **omnibus (table-level)**; cell claims rest on log2 effect sizes.
- Coverage assumes non-overlapping intervals within method (verified upstream).
- Class imbalance: mRNA dominates; diverse, ribosomal protein, and unknown are small — per-class n quoted on interpretation.

8. Data-integration validation

A staged join diagnostic (accession → +method → +start → +end → ±1 tolerance → interval overlap, per method) identified and corrected three integration defects:

1. **Method-label mismatch**: annotations FLPS/SEG vs. measurements `FLPS_strict/SEG_strict`; resolved by a canonical alias map on both sides.
2. **Accession extraction** from pipe-delimited composite IDs (text before the first |); initial parsing had silently broken the UniProt-length join.
3. **Multi-sheet measurements** (composition, distribution, co_occurrence): all three sheets now merged on interval keys.

Post-fix join QC: **100% measurement match and 100% UniProt-length match for all six methods**; exact interval equality 100% per method, confirming identical coordinate sets.

9. Outputs and organization

- Per-method results: `results/methods/<method>/{tables,figures}`.
- Cross-method consolidation: `results/summary/{tables,figures}` (master stacked tables, headline numbers, class × method pivots, signature-enrichment matrix, comparison heatmaps/bars).
- Deep-read extraction: `results/summary/deep_read/{tables,figures}` (charge-landscape summaries, significant coverage cells, concordance check).
- `RESULTS_GUIDE.md` and `README_deep_read.md`: reading order and question→output maps.

10. Findings

10.1 Detection methods fall into three archetypes (headline numbers)

Method	n LCRs	n proteins	Median LCR length	Median coverage	% multi-LCR	Mean LCRs/protein
AlcoR	2,804	968	25 aa	0.295	61.1%	2.90
CAST	1,880	1,055	76 aa	0.294	44.6%	1.78
FLPS	8,099	1,697	17 aa	0.166	80.8%	4.77

Method	n LCRs	n proteins	Median LCR length	Median coverage	% multi-LCR	Mean LCRs/protein
LCRFinder	6,371	1,008	6 aa	0.047	86.2%	6.32
SEG_intermediate	1,307	622	26 aa	0.075	48.2%	2.10
SEG	493	296	21 aa	0.044	35.5%	1.67

- **Long-window (CAST), short-motif (LCRFinder), and intermediate (FLPS, AlcoR, SEG_intermediate, SEG)** archetypes differ by an order of magnitude in call length and coverage.
- Cross-method comparisons are therefore meaningful only on **protein-level** quantities; LCR-level metrics compare different objects across methods.
- Method effects dominate RNA-class effects in scale; class structure is read *within* each method.

10.2 Validated cross-method results (protein level)

Consensus board — final (6/6 methods assessed):

Signal	Support	Status
Ribosomal proteins: single basic/Arg-rich LCR at a domain boundary; no linker/terminal/no-Pfam LCRs	5/5 powered methods (SEG underpowered, directional)	Headline result
pre-rRNA/snoRNA: basic + acidic LCRs	6/6	Validated
snRNA (and mRNA): SR/RS repeats; snRNA polyampholyte + linker tendency	6/6	Validated
mRNA: GS-rich, RG/RGG, aromatic-sticker, disorder-spacer composition; multi-LCR in short-call methods	5/6 (composition); multi-LCR method-dependent	Validated
ncRNA: ~40% of proteins lack Pfam annotation	6/6 (+0.94...+1.70)	Validated — most consistent result
ncRNA: aromatic-aggregation-prone LCRs	Strong cells in CAST and FLPS at both levels	Validated
tRNA: few, domain-intrinsic LCRs	5/6 (CAST null = window-length artifact)	Validated
diverse: hydrophobic/aggregation-prone LCRs	3/5 powered methods	Validated, secondary thread
tRNA: cation- π enrichment	contradicted at protein level	Rejected (short-window artifact)
Arg-specific (vs. generic basic) enrichment in nucleolar classes	3/4 composition methods	Supported; claim charge-driven basic/Arg-rich

Headline result — the ribosomal-protein LCR architecture. Across AlcoR, FLPS, CAST, LCRFinder, and SEG_intermediate, ribosomal proteins (and ribosome-biogenesis proteins more broadly) are domain-dense proteins carrying **exactly one dominant basic/Arg-rich LCR positioned at a domain boundary**: single-LCR enrichment in every method (up to 95.7% single

in CAST), domain_edge/domain_intrinsic enriched (CAST edge +1.54, 69.6% of proteins; SEG_intermediate intrinsic +3.24), with near-total exclusion from interdomain linkers (0–0.8% of proteins), distal termini (2–12%), and no-Pfam proteins ($\leq 2\%$).

Nucleolar axis (pre-rRNA, snoRNA). Both classes carry basic + acidic LCRs in all six methods. Acidic arm: acidic_region presence in 92.9% of SEG-detected pre-rRNA proteins (+1.89), 67.6% snoRNA in FLPS (+0.67). Basic arm: basic_enriched_region enriched in pre-rRNA and snoRNA in all methods (e.g., CAST +1.66/+1.71). The Arg-specific refinement holds in AlcoR/FLPS/CAST but not LCRFinder (6 aa peaks rarely satisfy the combined Arg rule) — the claim is charge-driven basic/Arg-rich.

Splicing axis (snRNA, mRNA). sr_rs_repeat_region enriched in snRNA in all six methods at protein level (+0.49...+1.26; 15–32% of proteins), with mixed_charge_polyampholyte support (FLPS +0.56, CAST +0.62) and an interdomain_linker tendency in snRNA (CAST +1.00, SEG_intermediate +0.77, FLPS +0.30), consistent with RS domains between RRM in SR proteins.

mRNA class. Compositionally distinct: gs_rich_neutral_region, rg_rgg_repeat_region, aromatic_sticker_region, disorder_rich_spacer_region enriched across methods (disorder-spacer in 76% of LCRFinder mRNA proteins), basic signatures depleted. Multi-LCR architecture appears only in short-call methods (FLPS 72% 3+, LCRFinder 80% 3+) — method-dependent, reflecting call granularity.

ncRNA class. The single most consistent cross-method result: ~30–44% of ncRNA-class proteins lack Pfam annotation, enriched in all six methods (+0.94, +0.98, +1.11, +1.01, +1.70, +1.56). aromatic_aggregation_prone_region enrichment is machine-confirmed strong at both levels in CAST (+2.34/+2.35) and FLPS (+1.50/+1.69). ncRNA-binding proteins emerge as a partially uncharacterized, LCR-driven class — a priority for follow-up.

tRNA class. Few (mostly single) LCRs per protein (4/5 powered methods), domain_intrinsic placement (AlcoR +0.72, FLPS +0.57, LCRFinder +0.82; CAST null attributable to 76 aa windows exceeding domain size). tRNA enzymes are domain-dense with small embedded LCRs.

diverse class. Hydrophobic, aggregation-prone LCRs enriched in three powered methods at protein level (FLPS +1.10, LCRFinder +1.00, SEG_intermediate +3.53), with elevated unmapped_physicochemical_properties (SEG_intermediate +2.94) — atypical LCRs partially outside the current taxonomy; flagged for taxonomy review.

unknown class. Heterogeneous top signatures (Gln-rich, hydrophobic, Pro-rich), elevated multi-LCR and distal-terminal LCRs; retained as the control/QC narrative, as designed.

10.3 Continuous charge landscape validates the categorical results

The threshold-based signature calls are backed by distribution-level differences in charge metrics (per class $\times$ method, KW-tested):

- **Ribosomal basic charge is distribution-level, not a threshold artifact:** median NCPR positive in all six methods (+0.082...+0.267; the only class positive everywhere); median frac_positive the class maximum in 5/6 methods (CAST 0.33, FLPS 0.29, LCRFinder 0.37, SEG_intermediate 0.34, AlcoR 0.17).

- **The nucleolar classes are bipolar:** pre-rRNA combines high `frac_positive` (CAST 0.24, second to ribosomal) with the highest `frac_negative` of any class in 4–5/6 methods, and near-zero median NCPR in long-window methods (CAST +0.05, FLPS +0.06) with maximal FCR (class maximum in 5/6, 0.28–0.81). Near-zero NCPR + maximal FCR = coexisting basic and acidic tracts, not neutrality.
- **SEG variants resolve the acidic pole:** SEG pre-rRNA median NCPR = -0.74 (FCR 0.81); SEG_intermediate = -0.50 . The earlier signature-level difference (SEG's top pre-rRNA call = `acidic_region` vs. CAST/FLPS = `basic_enriched_region`) is therefore a **stringency effect** selecting which pole of the same bipolar landscape is detected, not a method disagreement.
- **mRNA neutrality confirmed continuously:** median NCPR ≈ 0 in every method; FCR in the lower band — mRNA LCRs are genuinely the least charged population.
- **snRNA positive pole in SEG** (median NCPR +0.35): consistent with Arg within RS runs; SEG's strict windows isolate exactly those runs.
- **AlcoR is the only method with no class structure in FCR** (KW FDR 0.18) — expected, as it detects coiled-coil propensity rather than composition bias; its contribution is positional/structural.
- FLPS/LCRFinder show large within-class IQRs (bimodality from call granularity: many zero-charge micro-calls plus dense ones) — another reason protein-level summaries carry the interpretation.

10.4 Coverage and signature-occupancy statistics (KW-tested)

- **Coverage differences across RNA classes are significant in every method** (protein-level KW FDR: FLPS 6.4e-27, LCRFinder 9.9e-17, CAST 6.6e-06, SEG_intermediate 8.0e-05, SEG 7.2e-04, AlcoR 0.019). Ribosomal protein coverage is the class maximum in 4/6 methods (CAST tops at mRNA, FLPS at snoRNA); tRNA is the lowest/near-lowest in CAST, FLPS, and LCRFinder.
- **The ribosomal LCR occupies the largest share of its parent protein in all six methods at LCR level** (`coverage_per_lcr` class maximum: CAST median 26% of protein length, AlcoR 13%, SEG_intermediate 13%, FLPS 10%, SEG 8.7%, LCRFinder 2.4%).
- **Signature occupancy (property fractions among carrier proteins):**
 - Arg-rich contact regions cover a median 28% of protein length in CAST and AlcoR ribosomal proteins (vs. 2–5% elsewhere; AlcoR FDR 0.006, FLPS 0.129, FDR 1.3e-15); basic regions cover 13–28% in CAST/FLPS/SEG_intermediate ribosomal proteins. Combined with the single-LCR architecture and boundary placement: **one dominant basic LCR occupying a substantial fraction of the protein at a domain boundary** — an RNA-contact arm, not a small patch.
 - Acidic-region occupancy is the class maximum in pre-rRNA (FLPS 0.076, LCRFinder 0.026) and elevated in snoRNA.
 - SR/RS repeat regions occupy 12–14% of protein length in mRNA/snRNA carriers (SEG_intermediate, FDR 0.0075) — a substantial structural element.
 - Disorder-spacer *presence* is depleted in ribosomal proteins, but its *fraction among carriers* is the class maximum (CAST 0.12, SEG_intermediate 0.20): presence and

magnitude are distinct statements; report both.

- Rows with $n = 1-4$ proteins (e.g., FLPS gs_rich ribosomal $n=3$) are descriptive only and excluded from formal claims.

10.5 Machine-checked concordance validation

`concordance_check.csv` places LCR-level $\log_2$ (sequence-count and residue-count bases) beside protein-level $\log_2$ for every method $\times$ feature $\times$ class cell, with computed flags: `concordant` (same direction both levels), `discordant` (opposite non-zero directions), `strong_cell` (concordant and ≥ 2 -fold at both levels, $|\log_2| \geq 1$).

- **All headline cells are machine-confirmed:** ribosomal Arg-rich/basic (CAST strong: +2.67/+1.95 and +2.28/+1.59), domain-edge and domain-intrinsic placement (strong in CAST, SEG_intermediate, LCRFinder, SEG), distal-terminal and linker/no-Pfam exclusions (strong in 5/5 powered methods), nucleolar basic (CAST pre-rRNA/snoRNA strong), snRNA SR/RS (AlcoR strong +1.81/+1.45), ncRNA no-Pfam (strong in CAST and SEG_intermediate; concordant elsewhere).
- **Depletions are as consistent as enrichments:** basic/Arg-rich depleted in mRNA and snRNA; aromatic-sticker, GS-rich, and disorder-spacer depleted across the nucleolar classes; SR/RS and RG/RGG depleted in ncRNA, pre-rRNA, and tRNA. Classes systematically avoid each other's LCR types.
- **Discordant cells are predominantly near-zero sign flips** (sampling noise). One genuine pattern: FLPS ribosomal `basic_enriched_region` is LCR-level enriched (+1.05) but protein-level neutral (−0.06) — within-protein clustering of FLPS calls inflating LCR-level counts; a worked example of the pseudo-replication the two-level rule was designed to catch, cited in methods.
- **Calibration note:** the `strong_cell` bar is strict — several validated results are "concordant but not strong" because protein-level effects are genuinely smaller (e.g., AlcoR `domain_edge` $\times$ ribosomal +0.86/+0.47). Strong cells = headline claims; concordant cells = support layer.

10.6 Rejected hypotheses and method artifacts

- **tRNA cation- π enrichment:** large LCR-level $\log_2$ values (LCRFinder +3.44, SEG +4.36) trace to single short-window observations; protein-level tables flat in CAST and LCRFinder. Rejected as a short-window artifact.
- **CAST positional nulls:** 76 aa median windows cannot fit inside typical domains, suppressing `domain_intrinsic` calls and inflating boundary ambiguity (edge vs. intrinsic assignments depend on window length). Positional comparisons across methods must account for call length.
- **LCRFinder single-residue signatures** (glycine/serine-rich) are near-tautological at 6 aa median length; its motif-level signatures (SR/RS, RG/RGG) are its trustworthy contributions.
- **SEG extreme $\log_2$ values** derive from tiny windows and small counts; n quoted alongside throughout.

- **SEG strict's role** is the strictness control: patterns confirmed in SEG are high-confidence; its nulls are underpowered, not contradictory.
- **AlcoR** measures coiled-coil propensity rather than composition bias; its hydrophobic/domain-intrinsic signals read accordingly.

11. Outputs inventory (for documentation)

- `headline_numbers.csv` — per-method key figures (§10.1).
- `master_lcr_categorical.csv` / `master_protein_categorical.csv` — all enrichment tables, both levels.
- `master_lcr_continuous.csv` / `master_protein_continuous.csv` — all metric summaries with median/IQR/mean/SD/KW p/FDR.
- `median_coverage_by_class_method.csv`, `median_length_by_class_method.csv` — class × method pivots.
- `top_signature_per_class.csv` — strongest enriched signature per class per method (LCR level; table-level p).
- `charge_landscape_summary.csv` + `charge_landscape_median_<metric>.csv` — charge metrics per class × method with KW statistics (§10.3).
- `significant_coverage_cells.csv` — KW-tested coverage and property-fraction cells (§10.4).
- `concordance_check.csv` — machine-checked LCR ↔ protein concordance with strong-cell flags (§10.5).
- **Figures:** coverage/length/signature heatmaps (summary), NCPR/frac-positive median heatmaps (`deep_read`), per-method stacked bars, multi-panel fraction histograms.

12. Next steps

1. Detailed documentation/manuscript drafting from this record.
2. Optional per-cell Fisher-exact pass if cell-level significance claims are required for specific headline cells.
3. Follow-up analysis of the ncRNA (no-Pfam, aromatic-aggregation) and diverse (hydrophobic-LCR) classes — the two most novel threads.
4. Taxonomy review for hydrophobic/aggregation-prone LCRs given the unmapped enrichment in the diverse class.